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Regulating CEO Pay: Evidence from the Nonprofit Revitalization Act

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1 Big picture

- **Question.** Does regulation that weakens CEOs' ability to influence their own compensation improve pay efficiency and nonprofit outcomes?
- **Setting.** The New York Nonprofit Revitalization Act of 2013 required conflicted officers to be absent from deliberations and votes about their compensation.
- **Core design.** The paper compares New York nonprofits before and after the Act with nonprofits in other states, and then compares CEOs with non-CEO executives in the same nonprofit.
- **Main results.** CEO pay falls by about 2%, CEO hours rise by about half an hour per week, donations and volunteers rise, and pay-donation sensitivity increases.
- **Takeaway.** A governance rule that targets the pay-setting process can lower executive pay while increasing effort and stakeholder support.

2 Data and measurement

2.1 IRS Form 990 and Schedule J sample

The data combine IRS Form 990 and Schedule J filings from 2009–2018. Form 990 reports nonprofit accounting and performance variables. Schedule J reports compensation for executives earning over \$150,000. The final sample contains 14,111 nonprofits, 450,771 executive-year observations, and 84,421 CEO-year observations.

Interpretation. The data allow comparisons across organizations, across states, across years, and within the same nonprofit between CEOs and other executives.

2.2 CEO identification, compensation, effort, and performance

The baseline defines the CEO as the highest-paid officer and checks robustness using AWS electronic filings with job-title identification. Compensation outcomes include total pay, salary, bonus, perquisites, deferred pay, and nontaxable benefits. Effort is measured by reported weekly hours. Performance is proxied by contributions and grants, volunteers, revenue per employee, and administrative efficiency.

Interpretation. Donations and volunteers are useful stakeholder-response measures because nonprofit output is multidimensional and often unobservable.

2.3 Treatment variables and controls

NY_i identifies New York nonprofits. $Post_t$ identifies fiscal years after July 2014. Governance controls include conflict-of-interest policy, whistleblower policy, audit committee, board independence, family/business relationships, leverage, assets, and revenue. Industries are one-digit NTEE codes.

Interpretation. The Act changes the compensation process rather than mandating a pay level, so changes in pay and outcomes reveal whether the prior process was inefficient.

3 Model and empirical specifications

3.1 Nonprofit production and donor objective

The CEO chooses costly effort e_H or e_L , which affects productivity γ . Donors choose donations D to maximize charitable goods production net of donation cost and expected compensation:

$$\max_D X(\gamma, D) - D - E[w(\gamma, s)], \quad \frac{\partial X(\gamma, D)}{\partial D} = 1.$$

Interpretation. Higher productivity raises the marginal return to donations.

3.2 Independent versus compromised boards

An independent board maximizes donor utility and chooses a contract that induces high effort. A compromised board is influenced by the CEO and maximizes CEO utility subject to an outrage constraint. The optimal independent contract satisfies

$$w(\gamma_G, 1) = w_G, \quad w(\gamma_B, 1) = w_B, \quad w(\gamma, 0) = w_0,$$

$$U(w_G) - U(w_B) \geq \frac{e_H - e_L}{(p_H - p_L)/2}, \quad w_G > w_0 > w_B.$$

Interpretation. When the CEO influences pay, pay-performance sensitivity is weaker and effort is lower.

3.3 Empirical specifications

The CEO-only difference-in-differences model is

$$\log(CEOPay_{it}) = \gamma_i + \delta_t + \beta(NY_i \times Post_t) + X'_{it}\theta + \varepsilon_{it}.$$

The triple-difference model is

$$\log(Pay_{ijt}) = \gamma_{it} + \delta_{jt} + \zeta_{ij} + \beta(NY_i \times Post_t \times CEO_{ijt}) + \varepsilon_{ijt}.$$

Interpretation. The DDD design compares CEO pay changes to changes in other executives' pay within the same nonprofit.

4 Identification strategy

The identification uses the July 2014 New York policy shock, firm fixed effects, year fixed effects, and high-dimensional fixed effects in the DDD specification. Event studies show no clear pre-trends for CEO pay and no comparable decline for non-CEO executives. Robustness checks use neighboring states, CEO-industry-year fixed effects, title-based CEO identification, spell fixed effects, and CEO turnover tests.

Interpretation. The key assumption is that, absent the Act, New York CEO pay would have evolved like control-state CEO pay and like non-CEO executive pay within the same organization.

5 Tables and figures

5.1 Figure 1: Types of nonprofit organizations

Read: The figure shows the distribution of nonprofit types in the 14,111-organization sample.

Interpretation. The sample is sectorally diverse, making industry controls important.

Economic message. Nonprofit governance is not confined to traditional charities.

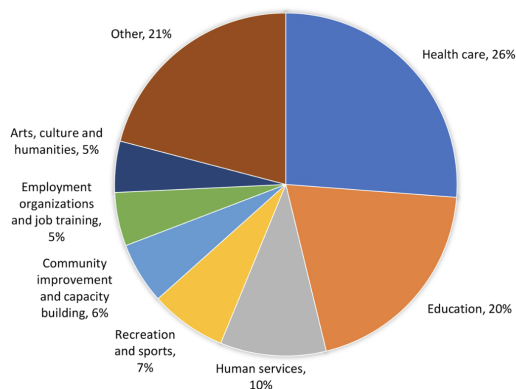


Figure 1
Types of nonprofit organizations.
This pie chart shows different categories of nonprofit organizations in the United States and their representation (in %) in our sample. The sample consists of 14,111 unique nonprofit organizations that filed form 990 with the Department of the Treasury Internal Revenue Service in the period 2009–2018.

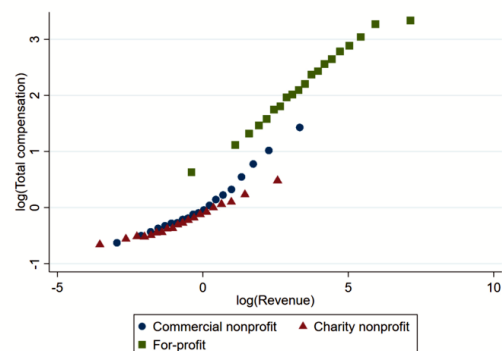


Figure 2
Total CEO compensation versus firm size.
This figure plots the relation between adjusted log total CEO compensation and adjusted log revenues for three types of organizations: for-profits (green squares), commercial nonprofits (blue circles), and charity nonprofits (maroon triangles). The methodology follows Gabaix and Landier (2008) and Gabaix (2016). We calculate the annual median CEO pay and revenue, then compute median-adjusted values. The data are grouped into 20 ventiles based on adjusted revenue, and we plot the average adjusted revenue against the average adjusted CEO

5.2 Figure 2: Total CEO compensation versus firm size

Read: CEO pay is strongly related to firm size; nonprofit CEOs earn less than for-profit CEOs; charities pay less than commercial nonprofits.

Interpretation. The figure motivates both size controls and the meaning-from-work mechanism.

Economic message. Some nonprofit CEOs may accept lower compensation because mission generates utility.

5.3 Table 1: Summary statistics

Read: The table summarizes compensation, firm characteristics, and pre-Act comparisons.

Interpretation. The table establishes the scale and richness of the sample.

Economic message. CEO pay in nonprofits is economically meaningful.

5.4 Table 2: Total pay of nonprofit executives

Read: CEO pay declines by about 1.6–2.4% after the Act.

Table 1
Summary statistics

A. All executives

	Mean	SD	p10	p50	p90	Obs.
Total compensation (\$000s)	405.46	362.35	168.24	275.28	787.56	450,771
Salary (\$000s)	277.85	188.86	134.21	212.13	507.39	450,771
Bonus pay (\$000s)	46.21	110.24	0	0.80	125.15	450,771
Perquisite pay (\$000s)	24.24	71.57	0	1.19	53.39	450,771
Deferred pay (\$000s)	30.06	53.26	0	14.88	64.42	450,771
Nontaxable benefits (\$000s)	17.79	13.63	2.08	16.13	32.50	450,771
CEO	0.12	0.32	0	0	1	450,771
Hours worked	43.22	6.60	40	40	54	450,771

B. CEOs

	Mean	SD	p10	p50	p90	Obs.
Total compensation (\$000s)	601.24	695.06	186.88	356.01	1269.43	84,421
Salary (\$000s)	367.75	268.52	155	276.83	709.86	84,421
Bonus pay (\$000s)	83.74	215.98	0	0	233.17	84,421
Perquisite pay (\$000s)	61.29	214.51	0	0.58	114.62	84,421
Deferred pay (\$000s)	47.68	107.53	0	15.6	110.13	84,421
Nontaxable benefits (\$000s)	20.10	19.98	0	16.18	40.44	84,421
Hours worked	43.48	7.08	40	40	55	84,421

C. Firm characteristics

	Mean	SD	p10	p50	p90	Obs.
Number of executives	5	5	0	4	15	84,421
Assets (\$M)	273.60	1559.38	10.97	61.50	454.42	84,421
Revenue (\$M)	130.48	602.65	5.19	28.60	257.01	84,421
Debt/Assets	0.38	0.33	0.04	0.31	0.80	84,418
Board independence	0.88	0.23	0.64	0.98	1	84,382
COI policy	0.94	0.24	1	1	1	84,421
WB policy	0.87	0.33	0	1	1	84,421
Audit committee	0.93	0.25	1	1	1	84,421
Family relation	0.22	0.41	0	0	1	84,421
Compensation committee	0.56	0.50	0	1	1	84,421
Compensation consultant	0.29	0.45	0	0	1	84,421
Employment contract	0.33	0.47	0	0	1	84,421
Review other firms' 990s	0.66	0.47	0	1	1	84,421
Independent comp. review	0.90	0.30	0	1	1	84,421
Charity	0.27	0.44	0	0	1	71,098
Female CEO	0.24	0.43	0	0	1	47,555
Contributions & grants (\$M)	14.16	41.22	0	1.75	29.98	84,421
Employees (hundreds)	8.84	17.02	0.20	2.71	22.98	84,421
Volunteers (hundreds)	5.98	21.83	0	0.50	9.50	84,421
Revenue per employee	0.33	0.77	0.04	0.11	0.62	84,421
Administrative efficiency	0.83	0.11	0.71	0.85	0.94	70,862
Labor expense per employee (\$000s)	61.50	47.31	18.75	49.75	116.22	84,421

D. CEO compensation and other variables before the Act

	NY State	Other states	p-value
Total compensation (\$000s)	626	621	.83
Compensation growth rate	0.024	0.027	.60
log(Assets)	4.254	4.223	.55
log(Revenue)	3.572	3.547	.64
Debt/Assets	0.378	0.373	.63
COI policy	0.946	0.951	.45
WB policy	0.900	0.891	.35
Audit committee	0.945	0.950	.50
Board independence	0.883	0.875	.48
Family relation	0.251	0.229	.15
	0.251	0.229	.15

This table reports summary statistics for the variables in our sample. The sample consists of 14,111 unique nonprofit organizations that filed form 990 with the *Department of the Treasury Internal Revenue Service* in the period 2009–2018. Panel A provides summary statistics for the full sample of executives (firm-year-executive level); panel B is for the sample of CEOs only (firm-year level). Panel C provides summary statistics for firm characteristics and governance variables (firm-year level). Panel D reports the means of the total CEO compensation, its growth rate, and of control variables for nonprofits located in the state of New York and in all other U.S. states. The means are calculated using data before the effective date of the *Nonprofit Revitalization Act of 2013*. p-values for the difference in means based on the standard errors clustered by firm are reported in the last column. Variable definitions are provided in [Appendix B](#).

Table 2
Total pay of nonprofit executives

<i>Executives:</i> <i>States:</i>	<i>log(Total compensation)</i>					
	<i>CEOs</i> <i>All</i> (1)	<i>CEOs</i> <i>All</i> (2)	<i>CEOs</i> <i>All</i> (3)	<i>All</i> <i>All</i> (4)	<i>All</i> <i>Neighbors</i> (5)	<i>All</i> <i>All</i> (6)
NY×Post	−0.017*** [−4.61]	−0.016*** [−4.56]	−0.016*** [−4.66]			
NY×Post×CEO				−0.020*** [−6.11]	−0.024** [−3.92]	−0.023*** [−6.47]
log(Assets)		0.089*** [9.45]	0.070*** [7.81]			
COI policy		0.002 [0.17]	0.002 [0.17]			
WB policy		−0.012* [−1.98]	−0.011* [−1.77]			
Audit committee		0.016** [2.27]	0.015** [2.12]			
log(Revenue)			0.039*** [7.20]			
Debt/Assets			0.026** [2.35]			
Board independence			−0.083*** [−3.79]			
Family relation			0.014** [2.10]			
Observations	84,421	84,417	84,220	450,771	125,547	450,767
R-squared	.931	.932	.932	.803	.825	.803
Firm FE	Y	Y	Y	N	N	N
Year FE	Y	Y	Y	N	N	N
Firm-year FE	N	N	N	Y	Y	Y
Firm-CEO FE	N	N	N	Y	Y	Y
CEO-year FE	N	N	N	Y	Y	N
CEO-ind-year FE	N	N	N	N	N	Y

The table reports the OLS estimates from the regressions, in which the dependent variable is the logarithm of total annual pay for nonprofit executives. *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. Specifications (1)–(3) use the sample of CEOs only; specifications (4)–(6) use the full sample of executives. In specification (5) the sample is limited to executives in the state of New York and neighboring states (PA, CT, MA, NJ, and VT); other specifications use data from all the states. The industries are defined by 1-digit NTEE codes (26 industries). *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

Interpretation. The result is stable across DD, DDD, neighboring-state, and industry-specific trend specifications.

Economic message. The Act reduced CEO pay without imposing a pay cap.

5.5 Figure 3: Total CEO compensation in nonprofit organizations

Read: The event study shows no clear pre-trend and a post-Act decline.

Interpretation. This supports the timing and parallel-trends argument.

Economic message. The pay decline follows the Act rather than preceding it.

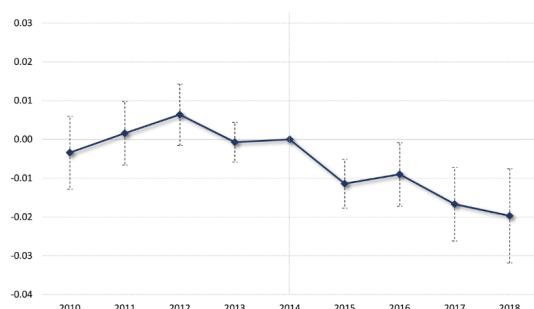


Figure 3
Total CEO compensation in nonprofit organizations.
This figure displays the OLS regression coefficients β_k and their respective standard errors, estimated from the following model: $\log(Pay_{ijt}) = \alpha_i + \nu_{jt} + \sum_{k=2009}^{k=2018} \beta_k \times NY_{it} \times D_k + \epsilon_{ijt}$, where the dependent variable is the logarithm of total CEO pay. The model includes firm and year fixed effects, variable NY_{it} is equal to one if a firm i is in New York in year t , and D_k is an indicator variable equal to one for observations in the fiscal year k . The sample includes CEOs of nonprofit organizations in the sample.

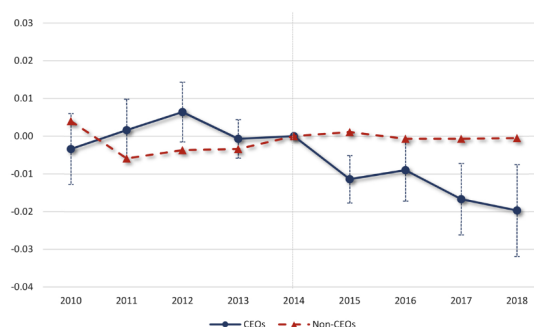


Figure 4
Total compensation for nonprofit CEOs and other executives.
This figure displays the OLS regression coefficients β_k and their respective standard errors, estimated from the following model: $\log(Pay_{ijt}) = \alpha_i + \nu_{jt} + \sum_{k=2009}^{k=2018} \beta_k \times NY_{it} \times D_k + \epsilon_{ijt}$, where the dependent variable is the logarithm of total executive pay. The model includes firm and year fixed effects, variable NY_{it} is equal to one if a firm i is in New York in year t , and D_k is an indicator variable equal to one for observations in the fiscal year k . The blue solid line represents the OLS estimates for the sample of nonprofit CEOs. The blue solid line represents the OLS estimates for the sample of other nonprofit executives (non-CEOs).

5.6 Figure 4: Total compensation for nonprofit CEOs and other executives

Read: CEOs show a post-Act decline; non-CEO executives do not.

Interpretation. This supports the CEO-specific treatment channel.

Economic message. The effect is not a general New York nonprofit labor-market shock.

5.7 Table 3: Components of nonprofit CEO pay

Read: Salary, bonuses, perquisites, and deferred pay decline; bonus extensive margins also decline.

Interpretation. Discretionary pay components fall the most.

Economic message. The Act changes both level and structure of compensation.

5.8 Table 4: Hours worked per week

Read: CEO hours increase by roughly 0.43–0.52 hours per week.

Table 3
Components of nonprofit CEO pay
A. Components of CEO pay

	<i>log(Salary)</i>	<i>log(Bonus pay)</i>	<i>log(Perquisite pay)</i>	<i>log(Deferred pay)</i>	<i>log(Nontaxable benefits)</i>
<i>Executives:</i>	<i>CEOs</i>	<i>CEOs</i>	<i>CEOs</i>	<i>CEOs</i>	<i>CEOs</i>
	(1)	(2)	(3)	(4)	(5)
NY×Post	-0.011*** [-2.79]	-0.034* [-1.71]	-0.073*** [-4.26]	-0.071*** [-2.98]	-0.015 [-1.20]
Observations	84,421	84,419	84,344	84,367	84,416
R-squared	.814	.752	.720	.702	.670
Firm FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
% Total pay	61	14	10	8	3

B. CEO Bonus and incentive compensation (extensive margin)

	<i>Bonus > \$25K</i>	<i>Bonus > \$50K</i>	<i>Bonus > \$75K</i>
<i>Executives:</i>	<i>CEOs</i>	<i>CEOs</i>	<i>CEOs</i>
	(1)	(2)	(3)
NY×Post	-0.012*** [-3.22]	-0.017*** [-3.92]	-0.010*** [-2.84]
Observations	84,421	84,421	84,421
R-squared	.695	.715	.728
Firm FE	Y	Y	Y
Year FE	Y	Y	Y

Panel A reports OLS regression estimates, where the dependent variable is one of the five reported components of annual compensation for nonprofit executives: *log(base compensation)*, *log(bonus and incentive compensation)*, *log(other reportable compensation and perquisites)*, *log(retirement and deferred compensation)*, and *log(nontaxable benefits)*. Panel B reports OLS regression estimates, with the dependent variables in columns 1-3 equal to one if CEO bonus and incentive compensation exceed \$25K, \$50K, or \$75K, respectively. *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. The sample includes CEOs of nonprofit organizations and all states. The last row in the table shows the average fraction of the specific component of executive pay in total executive pay. *t*-statistics are based on standard errors clustered by state and are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 4
Hours worked per week

	<i>Hours worked</i>					
<i>Executives:</i>	<i>CEOs</i>	<i>CEOs</i>	<i>CEOs</i>	<i>All</i>	<i>All</i>	<i>All</i>
<i>States:</i>	<i>All</i>	<i>All</i>	<i>All</i>	<i>All</i>	<i>Neighbors</i>	<i>All</i>
	(1)	(2)	(3)	(4)	(5)	(6)
NY×Post	0.427*** [6.56]	0.451*** [6.89]	0.444*** [6.74]			
NY×Post×CEO				0.465*** [5.16]	0.501** [2.49]	0.523*** [5.42]
log(Assets)		0.307* [1.68]	0.292 [1.60]			
COI policy		-0.461 [-1.24]	-0.511 [-1.39]			
WB policy		-0.437** [-2.22]	-0.410** [-2.05]			
Audit committee		0.772*** [4.95]	0.751*** [4.70]			
log(Revenue)			0.027 [0.29]			
Debt/Assets			-0.251 [-0.95]			
Board independence			-1.223*** [-3.25]			
Family relation			0.120 [0.84]			
Observations	49,909	49,904	49,729	175,362	44,796	175,351
R-squared	.819	.820	.820	.715	.742	.716
Firm FE	Y	Y	Y	N	N	N
Year FE	Y	Y	Y	N	N	N
Firm-year FE	N	N	N	Y	Y	Y
Firm-CEO FE	N	N	N	Y	Y	Y
CEO-year FE	N	N	N	N	N	N
CEO-ind-year FE	N	N	N	N	N	Y

The table reports the OLS regression estimates, in which the dependent variable is average hours worked per week by a nonprofit executive. *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. Specifications (1)-(3) use the sample of CEOs only; specifications (4)-(6) use the full sample of executives. In specification (5) the sample is limited to executives in the state of New York and neighboring states (PA, CT, MA, NJ, and VT); other specifications use data from all the states. The industries are defined by 1-digit NTEE codes (26 industries). *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

Interpretation. Lower pay is accompanied by higher effort.

Economic message. The evidence favors improved incentives over a simple talent-loss story.

5.9 Table 5: Firm performance: monetary donations and number of volunteers

Read: Contributions and grants rise about 4.3%; volunteers rise about 2.5–2.6%.

Interpretation. Stakeholder support rises after the Act.

Economic message. The pay reduction is accompanied by better performance proxies.

Table 5
Firm performance: Monetary donations and number of volunteers

	<i>Contributions & grants</i>			<i>Volunteers</i>		
<i>Sample:</i>	<i>Full</i>	<i>Full</i>	<i>Compliant with</i>	<i>Full</i>	<i>Full</i>	<i>Compliant with</i>
	(1)	(2)	COI, WB, Audit	(4)	(5)	COI, WB, Audit
			(3)			(6)
NY×Post	0.043*** [4.47]	0.043*** [5.05]	0.054*** [5.04]	0.026*** [4.87]	0.025*** [4.82]	0.024*** [3.34]
log(Assets)		0.315*** [12.15]			0.048*** [5.06]	
COI policy		-0.037** [-2.42]			0.001 [0.08]	
WB policy		0.016 [1.02]			0.039** [2.23]	
Audit committee		0.006 [0.59]			0.014* [1.79]	
Observations	84,421	84,417	63,507	84,421	84,417	63,507
R-squared	.938	.940	.938	.929	.929	.926
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

The table reports the OLS regression estimates. The dependent variable in columns 1–3 is the logarithm of one plus the dollar value of contributions and grants (in million), and in columns 4–6 it is the logarithm of one plus the number of volunteers (in hundreds). *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. Specifications (3) and (6) use the sample of nonprofits that were unaffected by other provisions of the Act (i.e., had an audit committee, conflict-of-interest policy, and whistleblower policy in 2013); all other specifications use the full sample of nonprofits. *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6
Pay-donation sensitivity

	<i>log(Total compensation)</i>			
<i>Donation variable timing:</i>	(1) Contemp.	(2) Lagged	(3) Contemp.	(4) Lagged
log(Contributions & grants)×NY×Post	0.024*** [4.04]	0.018*** [3.65]		
log(Contributions & grants)×NY	-0.003 [-0.61]	-0.005 [-1.01]		
log(Contributions & grants)×Post	0.002 [0.29]	-0.007 [-1.53]		
log(Contributions & grants)	0.007* [1.74]	0.014*** [3.47]		
log(Volunteers)×NY×Post			0.049*** [5.09]	-0.007 [-0.82]
log(Volunteers)×NY			0.003 [0.58]	0.015** [2.66]
log(Volunteers)×Post			-0.006 [-0.72]	0.008 [0.91]
log(Volunteers)			0.006 [1.17]	0.000 [0.02]
Observations	82,662	66,030	82,662	66,030
R-squared	.953	.961	.953	.961
Firm-Pre/Post FE	Y	Y	Y	Y
Year-Pre/Post FE	Y	Y	Y	Y

The table reports the estimates from OLS regressions, in which the dependent variable is the logarithm of total annual pay for nonprofit CEOs. *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. *t*-statistics based on standard errors clustered by the firm are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

5.10 Table 6: Pay-donation sensitivity

Read: Pay becomes more sensitive to contributions/grants and contemporaneous volunteers.

Interpretation. Contracts become more performance-sensitive after the Act.

Economic message. Governance-process regulation can change incentive design.

5.11 Table 7: Methods used to establish compensation

Read: Boards are more likely to use committees, consultants, contracts, peer 990 reviews, and independent compensation reviews.

Interpretation. The pay-setting process becomes more formalized.

Economic message. Removing CEOs from deliberation changes board behavior.

Table 7
Methods used to establish compensation

	Comp. Committee (1)	Comp. Consultant (2)	Employ. Contract (3)	Review Other 990s (4)	Independent Comp. Review (5)
NY × Post	0.018*** [3.91]	0.012*** [3.29]	0.018*** [4.33]	0.014** [2.24]	0.017*** [3.18]
Observations	84,421	84,421	84,421	84,421	84,421
R-squared	.849	.849	.849	.825	.796
Firm FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y

The table reports the OLS estimates, in which the dependent variable is one of five dummy variables indicating a particular method used by the firm to establish executive compensation or the sum of these five variables. The five methods include: compensation committee, independent compensation consultant, formal employment contract, reviewing other firms' 990 filings, and independent compensation review. *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8
Heterogeneity in the effect of the Act on CEO pay
A. Commercial nonprofits versus charities

	log(Total compensation)			
	(1) CEOs	(2) All	(3) CEOs	(4) All
Nonprofit type:	Commercial	Commercial	Charity	Charity
NY × Post	-0.019*** [-4.74]		-0.006 [-1.10]	
NY × Post × CEO		-0.028*** [-6.19]		-0.004 [-0.67]
Observations	52,008	320,564	19,090	85,289
R-squared	.929	.806	.917	.799
Firm FE	Y	N	Y	N
Year FE	Y	N	Y	N
Firm-year FE	N	Y	N	Y
Firm-CEO FE	N	Y	N	Y
CEO-year FE	N	Y	N	Y

B. Male versus female CEOs

	log(Total compensation)			
	(1) CEOs	(2) All	(3) CEOs	(4) All
Nonprofit CEO:	Male	Male	Female	Female
NY × Post	-0.037*** [-7.33]		0.002 [0.10]	
NY × Post × CEO		-0.024*** [-3.99]		0.007 [0.57]
Observations	45,014	249,149	21,796	72,473
R-squared	.927	.855	.950	.921
Firm FE	Y	N	Y	N
Year FE	Y	N	Y	N
Firm-year FE	N	Y	N	Y
Firm-CEO FE	N	Y	N	Y
CEO-year FE	N	Y	N	Y

The table reports the OLS estimates from regressions, in which the dependent variable is the logarithm of total annual pay for nonprofit executives. *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

5.12 Table 8: Heterogeneity in the effect of the Act on CEO pay

Read: Effects are stronger at commercial nonprofits and for male CEOs.

Interpretation. This matches the model's mission/meaning heterogeneity.

Economic message. The treatment matters most where compensation incentives are most important.

5.13 Table 9: Heterogeneity in the effect of the Act on hours and performance

Read: Effort and performance effects are stronger for commercial nonprofits and male CEOs.

Interpretation. Heterogeneity appears across pay, effort, and performance.

Economic message. The evidence supports the meaning-from-work channel.

Table 9
Heterogeneity in the effect of the Act on hours and performance

A. Commercial nonprofits versus charities

	Hours worked (1)	Contrib. & grants (2)	Volunteers (3)	Hours worked (4)	Contrib. & grants (5)	Volunteers (6)
Nonprofit: NY×Post	0.577*** [6.60]	0.024** [2.36]	0.024*** [4.58]	0.178 [1.49]	0.085*** [6.55]	0.014 [1.35]
Observations	26,130	52,008	52,008	10,171	19,090	19,090
R-squared	.830	.908	.917	.843	.934	.939
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

B. Male versus female CEOs

	Hours worked (1)	Contrib. & grants (2)	Volunteers (3)	Hours worked (4)	Contrib. & grants (5)	Volunteers (6)
CEO: NY×Post	0.523** [2.45]	0.078*** [5.86]	0.051*** [4.58]	-1.442 [-1.41]	-0.083 [-0.88]	0.062*** [3.46]
Observations	20,585	45,014	45,014	10,151	21,796	21,796
R-squared	.837	.945	.938	.772	.904	.964
Firm FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

The table reports the OLS regression estimates, in which the dependent variable in columns 1 and 4 is average hours worked per week by a nonprofit CEO. The dependent variable in columns 2 and 5 is the logarithm of one plus the dollar value of contributions and grants (in million), and in columns 3 and 6 it is the logarithm of one plus the number of volunteers (in hundreds). *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 10
CEO pay: Cross-sectional relations

A. Univariate comparisons

	Charity	Commercial	p-value	Female	Male	p-value
Total compensation	448.425	677.778	.01***	537.135	649.898	.01***
Comp. growth rate	0.030	0.026	.33	0.030	0.029	.86
log(Assets)	4.275	4.226	.20	4.368	4.558	.01***
Female CEO	0.269	0.210	.01***			
Charity				0.311	0.246	.01***

B: Cross-sectional regressions

	log(Total compensation)					
	(1)	(2)	(3)	(4)	(5)	(6)
Charity	-0.181*** [-13.93]	-0.193*** [-17.36]	-0.096*** [-8.95]			
log(Assets)		0.247*** [72.62]	0.120*** [22.71]		0.262*** [63.12]	0.121*** [17.76]
COI policy		0.023 [1.28]	0.009 [0.49]		-0.044 [-1.23]	-0.016 [-0.45]
WB policy		0.119*** [8.88]	0.104*** [8.02]		0.097*** [4.95]	0.083*** [4.43]
Audit committee		-0.009 [-0.67]	0.004 [0.33]		-0.021 [-0.99]	-0.006 [-0.29]
log(Revenue)			0.153*** [27.43]			0.172*** [23.76]
Debt/Assets			0.139*** [9.34]			0.131*** [6.78]
Board independence			-0.331*** [-15.87]			-0.333*** [-10.11]
Family relation			0.083*** [9.18]			0.068*** [6.40]
Female CEO				-0.127*** [-8.99]	-0.086*** [-7.78]	-0.064*** [-6.25]
Observations	86,198	86,194	85,993	48,122	48,122	48,071
R-squared	.279	.502	.554	.245	.499	.557
Ind-year FE	Y	Y	Y	Y	Y	Y

Panel A reports the means of the total CEO compensation, its growth rate, and firm size for charities and commercial nonprofits, as well as for nonprofits with male and female CEOs. Means are calculated using data before the effective date of the *Nonprofit Revitalization Act of 2013*. *p*-values for the difference in means based on the standard errors clustered by firm are reported below. Panel B reports the OLS estimates from regressions, in which the dependent variable is the logarithm of total CEO pay. The industries are defined by 1-digit NTEE codes (26 industries). *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in [Appendix B](#). * $p < .1$; ** $p < .05$; *** $p < .01$.

5.14 Table 10: CEO pay: cross-sectional relations

Read: Charity CEOs and female CEOs earn less in the cross-section.

Interpretation. The evidence is suggestive rather than causal, but aligns with the model.

Economic message. Mission and meaning are associated with lower pay.

5.15 Table 11: Total pay: identification of CEOs based on job titles

Read: The pay decline remains with title-based CEO identification.

Interpretation. The main result is not driven by highest-paid-officer misclassification.

Economic message. CEO identification robustness strengthens the baseline interpretation.

5.16 Table 12: Total CEO pay with spell fixed effects and CEO turnover

Read: Pay falls within spells, and turnover does not rise significantly.

Interpretation. The pay effect is not driven by replacing CEOs with cheaper hires.

Table 11
Total pay: Identification of CEOs based on job titles

Executives: States:	log(Total compensation)					
	CEOs All (1)	CEOs All (2)	CEOs All (3)	All All (4)	All Neighbors (5)	All All (6)
NY × Post	-0.022*** [-4.48]	-0.022*** [-5.08]	-0.023*** [-5.62]			
NY × Post × CEO				-0.018*** [-3.66]	-0.022** [-3.63]	-0.025*** [-5.43]
log(Assets)		0.094*** [11.94]	0.070*** [8.43]			
COI policy		0.007 [0.37]	0.005 [0.24]			
WB policy		0.002 [0.16]	0.003 [0.28]			
Audit committee		0.016 [1.43]	0.015 [1.37]			
log(Revenue)			0.054*** [6.40]			
Debt/Assets			-0.023 [-1.34]			
Board independence			0.005 [0.34]			
Family relation			0.008 [1.24]			
Observations	67,243	67,136	67,094	391,062	108,290	391,062
R-squared	.859	.860	.860	.761	.778	.761
Firm FE	Y	Y	Y	N	N	N
Year FE	Y	Y	Y	N	N	N
Firm-year FE	N	N	N	Y	Y	Y
Firm-CEO FE	N	N	N	Y	Y	Y
CEO-year FE	N	N	N	Y	Y	N
CEO-ind-year FE	N	N	N	N	N	Y

The table reports the OLS estimates. The dependent variable is the logarithm of total annual pay for nonprofit executives. The sample is based on an Amazon database of electronic filings (10,695 firms), and CEOs are identified by their job titles. *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. Specifications (1)-(3) use the sample of CEOs only; specifications (4)-(6) use the full sample of executives. In specification (5) the sample is limited to executives in the state of New York and neighboring states. The industries are defined by 1-digit NTEE codes. *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in Appendix B. * $p < .05$; ** $p < .01$.

Table 12
Total CEO pay with spell fixed effects and CEO turnover

	log(Total compensation)			CEO turnover		
	(1)	(2)	(3)	(4)	(5)	(6)
NY × Post	-0.021*** [-5.78]	-0.022*** [-6.53]	-0.023*** [-7.29]	0.003 [0.82]	0.003 [0.82]	0.004 [0.85]
log(Assets)		0.063*** [7.11]	0.044*** [4.83]		-0.026** [-2.28]	-0.016 [-1.25]
COI policy		-0.015 [-0.46]	-0.016 [-0.48]		0.016 [0.52]	0.016 [0.53]
WB policy		-0.007 [-0.92]	-0.005 [-0.75]		-0.014 [-0.70]	-0.014 [-0.73]
Audit committee		0.022** [2.27]	0.021** [2.29]		-0.002 [-0.08]	-0.001 [-0.05]
log(Revenue)			0.045*** [8.12]			-0.020** [-2.11]
Debt/Assets			-0.016 [-0.90]			0.052*** [2.72]
Board independence			-0.009 [-0.53]			-0.039 [-1.31]
Family relation			0.004 [0.85]			-0.005 [-0.44]
Observations	57,123	57,041	57,005	55,887	55,791	55,764
R-squared	.945	.946	.946	.246	.246	.247
Firm FE	N	N	N	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Firm-person-title FE	Y	Y	Y	N	N	N

The table reports the OLS estimates from regressions. The sample is based on an Amazon database of electronic filings of form 990 (10,695 organizations), and CEOs are identified by their job titles. The dependent variable in specifications (1)-(3) is the logarithm of total annual pay for nonprofit executives, and specifications include spell fixed effects (based on Firm-Person-Title). The dependent variable in specifications (4)-(6) is CEO turnover, which is equal to one if the CEO of the firm changes from the prior year and zero otherwise. *NY* is equal to one if a nonprofit is based in NY; zero otherwise. *Post* is equal to one if all or a majority of a firm-year occurs after July 2014 (the effective date of the *Nonprofit Revitalization Act of 2013*); zero otherwise. *t*-statistics based on standard errors clustered by state are reported in brackets. Variable definitions are provided in Appendix B. * $p < .1$; ** $p < .05$; *** $p < .01$.

6 Interpretive synthesis

The Act lowered CEO pay, increased effort, improved stakeholder support, and increased pay-donation sensitivity. The complete pattern is consistent with a model in which compromised pay-setting gives CEOs too much influence and weakens incentives. Heterogeneity by commercial versus charity nonprofits and male versus female CEOs is consistent with mission meaning mitigating the need for financial incentives.

7 Short summary

- The Act prevented CEOs from participating in deliberations about their own pay.
- CEO pay fell by roughly 2%.
- CEO hours, donations, grants, volunteers, and pay-donation sensitivity rose.
- Boards adopted more formal pay-setting methods.
- Effects were stronger at commercial nonprofits and for male CEOs.
- Robustness tests support the baseline interpretation.

8 Conclusion

The paper shows that a narrow governance-process rule can improve nonprofit contracting. Lower CEO pay and higher effort/performance proxies suggest that pre-Act CEO influence allowed inefficient pay-setting in at least part of the nonprofit sector. The key lesson is that regulation aimed at the compensation process, rather than compensation levels, can improve organizational outcomes when managers influence their own contracts.